Programming Java

Event Handling

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Contents

The Delegation Event Model

- Applet is event-driven.
- Delegation Event model: JDK 1.1 introduced

addTypeListener() Method, removeTypeListener() Method

public void addTypeListener(TypeListener el)
public void addTypeListener(TypeListener el) throws TooManyListenersException
public void removeTypeListener(TypeListener el)

addActionListener() Method, removeActionListener() Method

void addActionListener(ActionListener al)
void removeActionListener(ActionListener al)

actionPerformed() Method

void actionPerformed(ActionEvent ae)

http://java.sun.com/j2se/1.5.0/docs/api/java/awt/Color.html

Event Classes

EventObject Constructor

EventObject(Object src)

getSource() Method

Object getSource()
String toString()

AWTEvent Constructor

AWTEvent(Object source, int id)

getID(), toString() Methods

int getID()
String toString()

ComponentEvent Constructor

ComponentEvent(Component src, int type)

Primary Event Classes

ActionEvent
AdjustmentEvent
ComponentEvent
ContainerEvent
FocusEvent
InputEvent
ItemEvent
KeyEvent
MouseEvent
TextEvent
WindowEvent

Event Classes

getComponent() Method

Component getComponent()

isAltDown(), ... Methods

boolean isAltDown()
boolean isControlDown()
boolean isMetaDown()
boolean isShiftDown()

getModifiers() Method

int getModifiers()

MouseEvent Constructor

MouseEvent(Component src, int type, long when, int modifiers, int x, int y, int clicks, boolean triggersPopup)

getX(), getY() Method

int getX() *int getY()*

getPoint()

Point getPoint()

translatePoint() Method

void translatePoint(int x, int y)

getClickCount() Method

int getClickCount()

Event Listeners

xxxListener() Methods

void addMouseListener(MouseListener ml)
void addMouseMotionListener(MouseMotionListener mml)
void removeMouseListener(MouseListener ml)
void removeMouseMotionListener(MouseMotionListener mml)

mouseXXX() Methods

void mouseClicked(MouseEvent me)
void mouseEntered(MouseEvent me)
void mouseExited(MouseEvent me)
void mousePressed(MouseEvent me)
void mouseReleased(MouseEvent me)

mouseDragged(), mouseMoved() Methods

void mouseDragged(MouseEvent me)
void mouseMoved(MouseEvent me)

```
import java.applet.*;
import java.awt.*;
import java.awt.event.*;
/*
  <applet code="MouseEvents" width=300 height=300>
  </applet>
*/

public class MouseEvents extends Applet
implements MouseListener {

  public void init() {
    addMouseListener(this);
  }

  public void mouseClicked(MouseEvent me) {
    setBackground(Color.blue);
    repaint();
  }

  public void mouseEntered(MouseEvent me) {
    setBackground(Color.green);
    repaint();
  }

  public void mouseExited(MouseEvent me) {
    setBackground(Color.red);
    repaint();
  }

  public void mousePressed(MouseEvent me) {
    setBackground(Color.white);
    repaint();
  }

  public void mouseReleased(MouseEvent me) {
    setBackground(Color.yellow);
    repaint();
  }
}
```

Adapter Classes

Adapter Classes

Adapter Class	***Listener Interface***
ComponentAdapter	*ComponentListener*
ContainerAdapter	*ContainerListener*
FocusAdapter	*FocusListener*
KeyAdapter	*KeyListener*
MouseAdapter	*MouseListener*
MouseMotionAdapter	*MouseMotionListener*
WindoAdapter	*WindowListener*

```
import java.applet.*;
import java.awt.*;
import java.awt.event.*;
public class MouseAdapterDemo extends Applet {

  public void init() {
    setBackground(Color.green);
    addMouseListener(new MyMouseAdapter(this));
  }
}

class MyMouseAdapter extends MouseAdapter {
  MouseAdapterDemo mad;

  public MyMouseAdapter(MouseAdapterDemo mad) {
    this.mad = mad;
  }

  public void mousePressed(MouseEvent me) {
    mad.setBackground(Color.red);
    mad.repaint();
  }

  public void mouseReleased(MouseEvent me) {
    mad.setBackground(Color.green);
    mad.repaint();
  }
}
```


Inner Classes

```
import java.applet.*;
import java.awt.*;
import java.awt.event.*;
/*
  <applet code="MouseInnerDemo" width=100 height=100>
  </applet>
*/

public class MouseInnerDemo extends Applet {

  public void init() {
    setBackground(Color.green);
    addMouseListener(new MyMouseAdapter());
  }

  class MyMouseAdapter extends MouseAdapter {

    public void mousePressed(MouseEvent me) {
      setBackground(Color.red);
      repaint();
    }

    public void mouseReleased(MouseEvent me) {
      setBackground(Color.green);
      repaint();
    }
  }
}
```

Anonymous Inner Classes

```
import java.applet.*;
import java.awt.*;
import java.awt.event.*;
/*
  <applet code="MouseAnonymousDemo" width=100
height=100>
  </applet>
*/

public class MouseAnonymousDemo extends Applet {

  public void init() {
    setBackground(Color.green);
    addMouseListener(new MouseAdapter() {

      public void mousePressed(MouseEvent me) {
        setBackground(Color.red);
        repaint();
      }

      public void mouseReleased(MouseEvent me) {
        setBackground(Color.green);
        repaint();
      }
    });
  }
}
```

An Anonymous Inner Class